Research Project Report

Politecnico di Milano · PhD Programme in Information Technology · 42nd cycle · Call 5722-5460

Area Research Area n. 1 - Computer Science and Engineering
Thematic research field Auditing and Mitigating Epistemic Failures in Web-Connected AI Agents (Pierri, Brambilla)
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Date September 2026

Project title Controlled evidence environments for auditing and mitigating provenance failures in web-connected agents

1. SUMMARY OF THE RESEARCH PROJECT

Problem statement, main expected results and originality. Max 500 characters including spaces and punctuation.

Web-connected agents launder evidence and omit sources, but live-web audits cannot say why: the auditor never knows what the agent could have retrieved. I will build a provenance-controlled environment, served to a pinned agent through a proxy and local index, and vary how far retrievable evidence sits from its origin (original, copy, paraphrase, AI rewrite) with model, harness and prompt fixed. Outputs: causal estimates for laundering and omission, a released environment, a validated gate.

2. DETAILED DESCRIPTION OF THE RESEARCH PROJECT

Emphasising original and innovative aspects and scientific relevance. Min 4,000 and max 8,000 characters including spaces and punctuation, excluding bibliography and pictures.

Evidence laundering and source omission, two failures named in this call, are failures of provenance: the agent presents derivative or synthetic material under an original's authority, or drops a source that contradicts its synthesis. Allaham and Diakopoulos found about sixteen percent of sources cited by generative search engines appear AI-generated; DeepTRACE found these systems routinely make statements their sources do not support. Neither can say why: on the live web the auditor does not know what the agent could have retrieved. Shah and Ozgur (2026) serve agents a generated mini-internet with credibility and factuality labels and plant one false article at a chosen rank; Yu, Kim and Kim (2026) dose a retrieval pool with AI-generated documents. Both control whether a source is credible or true. Neither controls how far a true claim sits from its origin: original, syndicated copy, paraphrase, AI rewrite. That derivation axis is what laundering is, and the Yu design has no browsing agent at all.

The design is an interception layer, not offline replay. The agent runs a search-and-fetch harness whose search tool queries a local OpenSearch index over a versioned corpus, and whose fetch tool passes through an HTTPS proxy that serves corpus documents under seeded hostnames and blocks anything else. Hostnames are subdomains of one registered domain under a public wildcard certificate: no trust store touched, no real publisher impersonated. Weights, harness commit and prompt are unmodified; DNS and proxy are the only configured parts. Each document carries a provenance record fixed at insertion: source identity, derivation position, factual status tied to a human-validated claim set, an AI-generation flag and a controllable timestamp. Claims come from post-cutoff and synthetic material and every item gets a closed-book baseline, separating memorised recall from retrieval. Year one targets 20,000 documents across 300 domains, half mirrored from Common Crawl, half derivatives from pinned generators. AI rewrites are cleaner and more keyword-dense than their originals and would outrank them, so the dose is set on the served result list, planted documents stratified across ranks, and the retriever (analyser, field weights, k, deduplication) is a declared, versioned factor. The harness is a pinned open-source ReAct-style scaffold, with a second scaffold as a replication arm. Models are open-weight and pinned locally; frontier models by API are a secondary arm.

Study A, evidence laundering: I plant pairs, an original of known source and derivatives that repeat its claims under altered or synthetic provenance, and vary the share of derivatives in served results (0, 10, 30, 50 percent). Each derivative carries a marker the original lacks, a unique figure or name, so marker-in-answer with citation-to-original is laundering by construction. Study B, source omission: I plant contradicting documents from sources of controlled quality, vary the support balance, and score omission and one-sidedness on the DeepTRACE dimensions. The unit of analysis is the claim-citation pair, with a random intercept per trajectory. An LLM judge, whose reliability for evidence-based agents is itself in question (Wang et al.), is validated against 1,500 double-annotated pairs at a target Cohen's kappa of 0.75, with a 100-pair pilot and a third rater on disagreements. This, not the GPU, is the dominant cost: two trained annotators, about 150 hours, roughly 2,500 euros over two years, with Prolific through the group's IRB route as the fallback if kappa falls short. Judge error is corrected by prediction-powered inference: the estimate from the judge-labelled set is debiased by a rectifier, the mean human-judge discrepancy on a random labelled subset (Angelopoulos et al.); Egami's design-based estimator is the robustness check.

The mitigation is a provenance-aware retrieval gate: a small encoder classifier placed between retrieval and synthesis. It scores each retrieved document on derivation position, AI-generation likelihood and source identity, then annotates it for the synthesiser or withholds it. This is weak-model oversight: the gate is far smaller than the agent it checks, and its error rate is reported against the same human reference set, not assumed. To keep it off labels that are true by construction, it is tested on derivatives from a held-out generator, against credibility re-ranking baselines, on reduction in laundering and omission, robustness to adversarial derivatives written to pass it, latency and cost, and usefulness rated by humans on held-out tasks.

The core is the environment, Study A and the gate. Year one: environment, corpus and claim set, pilot annotation, Study A in English. Year two: full annotation, gate, second scaffold and frontier arms, dataset release. Year three: usefulness study, thesis, and, if year one shows an effect, Study B, the adversarial arm, an Italian replication and the six months abroad. A subset of Study A tasks also runs on the live web, to check that environment rates fall in the live range, and with the retriever swapped, to check whether a rate survives an environment swap.

This environment is what I build now. At M+C Saatchi Fluency I run 68 collectors over ten platforms daily with an automated repair layer; at Dubizzle Labs I led 500 collectors across fifteen countries; I maintain web-scraping-guide.com, on anti-bot systems and agentic browsers. My one research artefact is an unpublished manuscript on cross-platform measurement equivalence, written with Ella Haig: a 500-replication simulation in which a pooled index reports the wrong sign on roughly a quarter of days. What I do not bring, I name: I have not fine-tuned a generative model, built a multi-turn agent or run a human-subjects experiment. The studies are sized to that; the harness is borrowed, not built. Annotation and rating go to the PoliMi ethics committee; I have been through this once at Portsmouth (TETHIC-2025-111094).

3. MOTIVATIONS

Reasons that led to wanting to carry out research at Politecnico di Milano. Max 2,000 characters including spaces and punctuation.

Francesco Pierri's Communications of the ACM article calls the integration of LLMs into search and social platforms a regulatory blind spot that should be urgently addressed, and this call is that sentence turned into a programme.

The group's Chinese search audit says in its limitations that annotation relies partly on LLM-assisted labelling, checked on samples, and that errors may remain. I have met that problem. At ArtemisAI our classifier scores rose from 71.6 to 85.8 across two versions, and part of that rise was the reference set moving from a human holdout to a model's labels. In industry there is no budget for stopping to ask what the number means. A lab that names its measurement problem in print is a lab I can be useful to. In return it supplies what I cannot supply myself: a decade of audit method, IRB-approved Prolific experiments, and venues where a finding about the web is what counts.

Marco Brambilla's lab lists data extraction and scraping among its interests and has recent work on LLMs inside graph-based RAG architectures. The environment I propose is a retrieval architecture with a provenance layer; I want it examined by someone for whom that is a research object rather than plumbing.

I have spent seven years building the instruments this call audits. A PhD is the only setting where holding the environment fixed for three years is itself the job.

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DECLARATION ON THE USE OF AI

Not counted toward the character limit.

AI use (Claude, Anthropic) was limited to draft critique, reference checking and condensing text to the character limits. The research design, claims and sources are my own and were verified by me.